Optimizer Trajectories Mislead:
Gradient Decomposition Recovers Feature Directions in Multitask Training

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Core idea. Most interpretability work that analyzes training dynamics assumes: *directions extracted from optimizer trajectories reflect feature formation.* This is wrong in a precise and fixable way.

Setup.

- **Task:** grokking on modular arithmetic ($a+b$, $a-b$, $a\cdot b$, a^2+b^2 mod 97).
- **Probe:** the Linear Centroids Hypothesis (LCH) of Walker et al. [**?**] — features as linear directions of input-space centroids $\mu_x = \nabla_x \ell_y(x)$, with sensitivity ratio $R_k(t)$ = centroid response along direction v_k vs. a random direction.
- **Compare two objects:**
  - Update SED: SVD over rolling AdamW updates $\Delta\theta_t$.
  - Gradient SED: SVD over rolling gradients $g(t) = \nabla L|_{\theta_t}$.

Result 1 — Single-task.

- Update SED: weak coupling, $\bar{R}_k \in [3, 9]$.
- Gradient SED: strong coupling, $\bar{R}_k \in [100, 330]$ across 4 ops and 3 seeds.

But a W ablation (window size) reveals: R_1=648 at W=1, decaying monotonically to 140 at W=40. The signal is dominated by instantaneous gradient alignment, which is near-tautological — $\nabla_\theta L$ and μ_x are built from the same Jacobians of the network. *Rolling-window SVD acts as denoising, not as the source of the coupling.*

Result 2 — Multitask (the load-bearing finding). A multitask transformer with shared encoder and 4 op-specific heads, trained on all four ops jointly:

- Update SED *collapses*: $\bar{R}_k \leq 1$ across all 4 ops — worse than random.
- Per-task gradient SED *recovers signal*: $\bar{R}_k \in [20, 45]$, consistent across all 4 ops.

The aggregated AdamW update mixes (i) momentum, (ii) adaptive scaling, (iii) weight decay, and (iv) inter-task gradient cancellation, destroying alignment with per-task feature directions. The fix: decompose gradients by task, then SVD per task. This requires only one extra backward pass per task per measurement checkpoint — cheap relative to training.

Result 3 — Causal intervention. Constraining the AdamW attention update to a rank-3 subspace, 3 seeds, addition and multiplication (15+15 = 30 runs):

mode (gradient-projected)	speedup over control
remove SED rank-3	1.02× (no effect)
keep only SED rank-3	**2.20**×
remove random rank-3	1.00× (no effect)
keep only random rank-3	**2.36**×

The acceleration is **rank-specific, not direction-specific**. Any rank-3 keep yields ≈ 2.3× speedup; SED-vs-random is within seed variation. So the SED basis is *diagnostic* of where centroid feature formation is concentrated, but *not causally privileged*: the rank constraint does the work.

Takeaways.

1. Optimizer trajectories ≠ feature directions.
2. Gradient structure is the right object.
3. Multitask requires per-task gradient decomposition.
4. Low-rank structure matters more than specific directions for whether grokking happens; specific directions matter for measuring *where* it happens.

One-line punchline. *If you want to understand what a network is learning, look at gradients — not what AdamW did to them.*

Why this matters.

- Explains apparent failures of trajectory PCA / SVD diagnostics in multitask settings.
- Connects to multitask gradient-interference literature.
- Suggests a candidate diagnostic for monitoring per-task feature formation in instruction-tuning, MoE, and other shared-encoder pipelines, where gradient interference is pervasive — though we have only verified the claim on a toy modular-arithmetic setting and large-scale validation is open.

Full paper, code, and per-checkpoint data caches: `https://github.com/skydancerosel/grokking-integrability`

References

[1] Walker, Humayun, Balestriero, Baraniuk. *The Linear Centroids Hypothesis: How Deep Network Features Represent Data.* arXiv:2604.11962 (2026).